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Key Points:

- We develop an algorithm to progress ice observations over inland water bodies by improving surface classifications using data derived from ICESat-2, Landsat 8/9 and other operational products, using the Laurentian Great Lakes as a testbed for large lakes on a global scale
- Two prediction data driven machine learning methods were developed that outperformed the current operational classification method in place for ICESat-2 Inland Water Surface (ATL13) by 26.46% and 20.37%
- Improved surface classification over large lakes refines data sets of ice concentration, freeboard and thickness for large lakes, allowing for more informative decision making, public safety, and climate change monitoring

Supporting Information:

Supporting Information may be found in the online version of this article.

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A Novel Algorithm for Ice-Water Discrimination in Large Lakes Using ICESat-2 Altimetry and Data Driven Machine Learning

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Abstract Large freshwater lakes are critical for human life, ecosystem functioning, and the global carbon cycle. However, consistent high-resolution methods to characterize ice over large lakes remain limited. Here we develop an algorithm to progress ice observations over inland bodies of water by improving surface classifications using data derived from ICESat-2, Landsat 8/9 and other operational products. This algorithm implements a hierarchical approach composed of remote sensing products and data driven machine learning. In this study we show that although the current classification method used in ICESat-2 Inland Surface Water Height (ATL13) is prone to overgeneralization and misclassification, our proposed algorithm, which integrates novel classification methods and data-driven machine learning, enhances surface classification accuracy. We tested this algorithm on a wide breadth of data, spanning four ice seasons in the Laurentian Great Lakes. In our algorithm, we developed two prediction methods that outperformed the current classification method in place for ATL13 by 26.46% and 20.37% and is scalable to other inland surface waters because of the global coverage of the necessary parameters for surface classification. Improved surface classification allows for inland surface bodies of water to be observed with greater detail, particularly using ICESat-2 data, and enables the production of improved data sets of ice concentration and thickness. Improved ice information on Earth's largest lakes will have cascading effects on not only public safety and operational efficiency, but also the monitoring of anthropogenic changes in these bodies of water.

1. Introduction

More than 84% of Earth's liquid surface freshwater exists in 10 of the largest freshwater lakes; yet, high resolution models of ice cover are limited to only a subset of Earth's large lakes, where ice predictive accuracy is limited or unknown (Anderson et al., 2018; Shiklomanov, 1993). The use of remote sensing has allowed for cryospheric research to grow extensively in extreme environments, but the implementation of these methods for mid-latitude large lakes has just scratched the surface. Not only do large freshwater lakes play a crucial role in supporting human life, they also serve as integral components of ecosystem functioning and the global carbon cycle (Williamson et al., 2009). Large lakes have been referred to as the “sentinels of change” such that they can be used to observe the effects of climate change, similar to how we observe changes in the ocean (Williamson et al., 2009). Having the ability to observe changes in ice characteristics will not only provide insights into how the climate is changing but also foster greater confidence in decision-making regarding human activities. For example, the commercial shipping industry transports more than 160 million tonnes of waterborne cargo every year via the Laurentian Great Lakes, hereafter the Great Lakes, and these shipping routes are affected by the presence of ice (Parran, 2020). Knowing the location and thickness of ice will enable more effective decision-making regarding ice-cutting operations, which would facilitate increased shipping along these routes. In addition to ice cutting operations, coastal ice conditions play an important role in coastal erosion, where sediments are often entrained and transported by winter wave energy, which impacts both human safety and ecological change (Barnes et al., 1994). Furthermore, the safety of people participating in recreational activities on ice covered lakes can be greatly improved if the quantification and characterization of the ice becomes more extensive, particularly in large lakes where ice extent has a high degree of temporal and spatial variability. In the Great Lakes, every year the US Coast Guard must be deployed to rescue civilians stuck on ice floes. The knowledge of where the ice may fracture would aid the Coast Guard in determining which areas of the lake people can access safely, thereby helping to mitigate ice-related safety risks (Kubat et al., 2021).

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In the Great Lakes, the National Oceanic and Atmospheric Administration (NOAA) CoastWatch Great Lakes Regional Node generates an Ice Cover Classification (ICECON) product largely based on synthetic aperture radar (SAR) to classify ice types (Leshkevich & Nghiem, 2013). Integration of multiple SAR missions into this product results in variable temporal resolution on the scale of days. At present, the only regional, daily product for ice observations over the Great Lakes are the Great Lakes Concentration Ice Charts produced by the U.S. National Ice Center and Canadian Ice Service, collectively known as the North American Ice Service (NAIS) (<https://usice-center.gov/Catalog/GlakesDailyCharts>). These charts are derived from a number of different sources (e.g., MODIS, GOES, AVHRR, Radarsat-2, Envisat) and are produced daily by the NAIS. Ice concentrations from this product are useful to stakeholder groups and model developers, but their resolutions are not sufficient for observing fine-scale ice characteristics. In addition to these product resolution limitations, both products are limited to the Great Lakes, and do not cover Earth's other large lakes, thus algorithm training, testing and validation are restricted to the Great Lakes to serve as a testbed.

Ice, Cloud, and land Elevation Satellite (ICESat-2) is a laser altimetry mission that has been gathering high accuracy information about the Earth's surface since its launch in 2018 (Markus et al., 2017; Neumann et al., 2019). The use of ICESat-2 over large inland lakes can lead to validation and improvement of high resolution ice prediction (Song et al., 2023). The ICESat-2 mission was designed to produce highly detailed observations of ice sheets, glaciers, sea ice, and vegetation, yet, its utility extends far beyond these categories (Markus et al., 2017). One of the objectives of the ICESat-2 mission was to determine the sea ice freeboard to an uncertainty less than 3 cm in the Arctic and Southern Oceans, which is particularly relevant to this study because ICESat-2 collects the same elevation data over many large ice-forming lakes like the Great Lakes, so high accuracy freeboard observations using ICESat-2 should be attainable (Markus et al., 2017). The Advanced Topographic Laser Altimeter System (ATLAS) aboard ICESat-2 has high vertical accuracy and collects data over the Great Lakes nearly every day, making it an ideal data source for studying ice in this region (Magruder et al., 2021). The ICESat-2 ATLAS Inland Water Surface Height Product (ATL13) has the potential to quantify lake ice characteristics like freeboard or thickness, but the underlying ice/water characterization system is insufficient for calculating these parameters (Jasinski et al., 2023). The surfaces from ATL13 are classified using the Daily Global Multi-sensor Automated Snow and Ice Map (GMASI) wherein GMASI classifications are projected onto the ICESat-2 data (Jasinski et al., 2023; Romanov, 2017). Misclassification of ATL13 data is likely a result of the coarse spatial resolution of GMASI (4 km grid cells) (Romanov, 2017). Surface classification accuracy is decreased because ice fields in the Great Lakes vary on a smaller scale than what the GMASI product can resolve (Wang et al., 2012). Low accuracy surface classification, regardless of the source, has cascading effects on the accuracy of ice observations like ice concentrations, water surface calculations of ice-covered regions, and ice-thickness calculations (Petty et al., 2021).

Ice/water discrimination algorithms developed for sea ice have the potential to be viable for inland lake ice, but these algorithms are designed for and use static threshold values for parameters like surface photon rates, background photon rates, and widths of the photon distributions, that are specific to sea ice (Kwok et al., 2022). In addition, the existing ICESat-2 Sea Ice Products (ATL07/10/20/21) algorithms do not produce data over lake ice regions, and implementation of data production in these regions would require additional algorithm modifications as well. The ICESat-2 Sea Ice Products implement a decision tree that uses static threshold values for surface classification that are not applicable to inland lake ice due to the seasonal and geographic dependencies of these parameters (Kwok et al., 2022). Additionally, the sea ice product algorithm is not presently able to use lead returns from low reflectance surfaces (Kwok et al., 2021), which are likely to be prevalent in lake ice. Although alternative approaches, like those developed for land-snow discrimination using photon background noise data (Zhang et al., 2020), and for sea ice-water (Koo et al., 2023; Zhang et al., 2019), have enhanced surface classification techniques, their application to large midlatitude lakes has not been directly investigated. Koo et al. (2023) developed an algorithm that integrated data-driven machine learning to classify ICESat-2 Sea Ice Product data, eliminating the dependence on static thresholds (Koo et al., 2023). The method, trained on 17 ICESat-2 tracks covering the Ross Sea, successfully categorizes sea surface types, but it has not been applied to or tested on lake ice. In this sea surface classification neural network, one of the implemented input parameters of the algorithms is a height relative to a local minimum, which allows for some indication of freeboard and therefore an ice surface. In theory, this parameter should work for inland lake ice, lake ice thickness is often on the scale of tens of centimeters, whereas sea ice can range from tens of centimeters to multiple meters thick (Golden et al., 2020; Volpano et al., 2020). When considering average ice thickness as well as the smaller freeboard values of lake ice

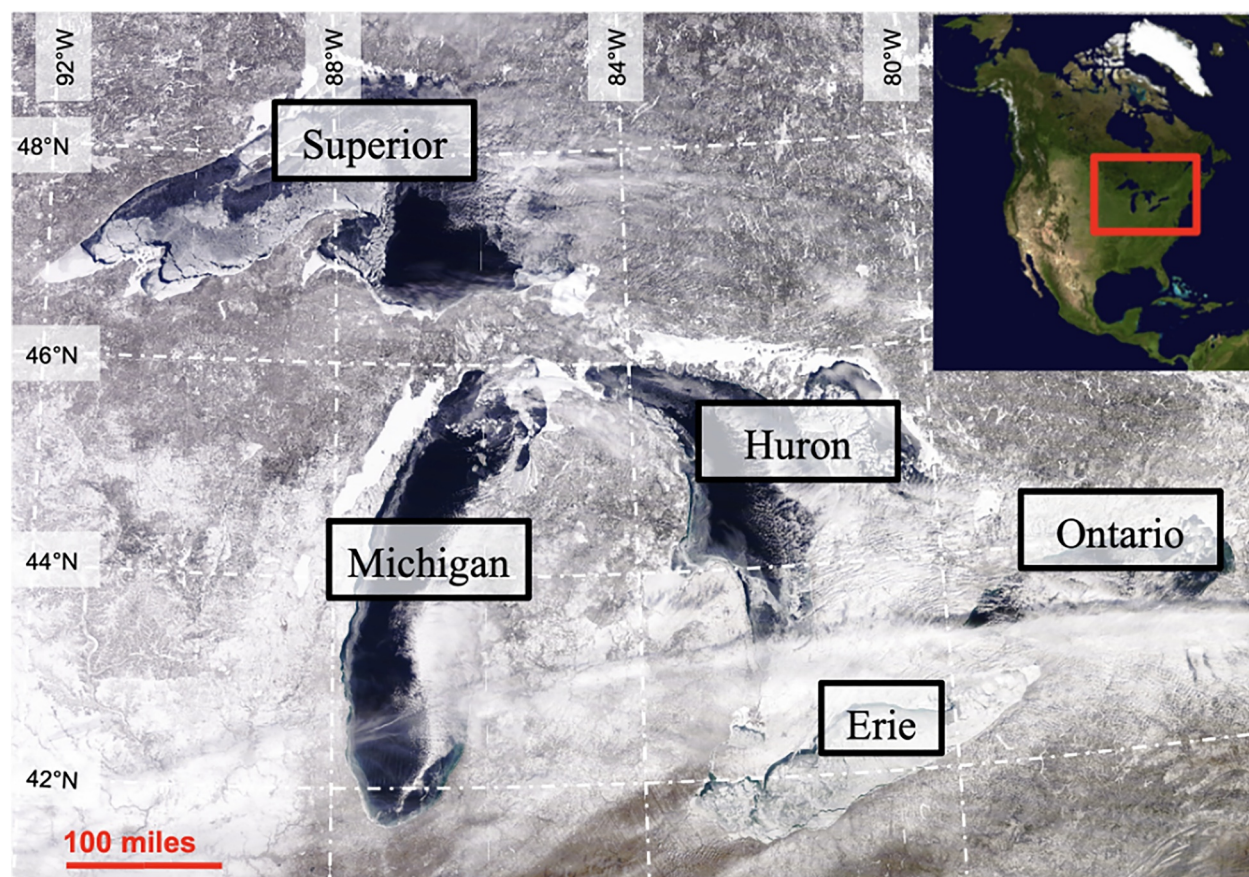


Figure 1. Moderate resolution imaging spectroradiometer image of the great lakes 02/19/2019 (<https://coastwatch.glerl.noaa.gov/satellite-data-products/modis/>).

due to the difference in water density, the relative height value could hold less weight in surface classification for freshwater lakes relative to sea ice. Additionally, some parameters of the sea surface classification algorithms are only available in ICESat-2's sea ice products, and not in ATL13, which would make application of these algorithms more difficult and would require use of the raw photon data from ICESat-2's ATL03 data product.

To obtain more reliable and consistent lake ice surface classifications using ICESat-2, data-driven machine learning can be implemented, as demonstrated by Koo et al. for sea ice surface classification. In this study, we develop an algorithm for inland freshwater that improves upon existing inland lake surface classification in ATL13. Although other methods of machine learning are available, we chose to integrate feedforward fully connected neural networks (NN) into our algorithm because they are well-suited for handling high-dimensional data sets and recognizing non-linear relationships within the data. We employ ICESat-2 data, Landsat 8 and Landsat 9 data (Landsat 8/9), and NOAA ICECON products for the years 2019–2023, wherein we use the Great Lakes as a testbed for this approach.

2. Study Area

We focus this study on the Great Lakes (Figure 1), which consist of Lakes Erie, Huron, Michigan, Ontario and Superior and contain 22.1% of the world's surface freshwater (Evans, 2004).

These lakes have been studied extensively and have substantial amounts of available data relative to other large freshwater lakes, which allows for more comprehensive cross validation of our methods (McBean & Motiee, 2008; Mortsch & Quinn Frank, 1996; Wang et al., 2012; Yang et al., 2020). These mid-latitude lakes straddle the United States–Canada border and span latitudes 41–49°N and longitudes 75–95°W. Some of Earth's largest lakes, like the Great Lakes, experience seasonal ice formation that can undergo high degrees of spatial and temporal variability due to the dynamic nature of these bodies of water. The ice conditions of the Great Lakes vary

spatially, from lake to lake, because of significant differences in depth, orientation, topography, and atmospheric conditions from lake to lake (Wang et al., 2012). For instance, Lake Erie experiences high concentrations of ice in most years even when Lake Ontario undergoes very little ice formation. Lakes Superior, Michigan, and Huron often fall somewhere in between in terms of ice concentration, with significant differences in extent and timing. The ice conditions of the Great Lakes vary spatially, from lake to lake because of each lakes orientation, depth, and turbidity (Wang et al., 2012). Large interannual fluctuations of ice coverage on the Great Lakes are frequently observed as well. For example, the maximum ice coverage was observed to be 56% for the 2021–2022 ice season and 23.35% for the 2022–2023 ice season (U.S. National Ice Center). This variability, especially in contrast to smaller lakes, creates complex ice-covered environments and allows these lakes to serve as critical case studies for understanding large lake ice dynamics.

3. Data

3.1. ICESat-2 ATL13 Inland Surface Water Data Product

ICESat-2 carries the ATLAS laser, which functions at a wavelength of 532 nm and is optically diffracted into six beams—three strong and three weak (Jasinski et al., 2023). These beams provide detailed surface elevation data with a vertical accuracy of approximately 5–8 cm (Jasinski et al., 2023). However, for lake ice freeboard studies the vertical measurement precision is more applicable to the freeboard uncertainty achievable. The vertical precision has been found to be ~1–2 cm over sea ice leads (Farrell et al., 2020). The satellite is in a 91-day repeat track orbit. The primary data set used in this study is the ATL13 Inland Surface Water Height product. The current ice/water classification system in place for ATL13 is GMAI, which has 4 km resolution. ATL13 provides height measurements for inland water bodies including lakes, rivers, estuaries, bays, and reservoirs. Data in ATL13 are aggregated into short segments, where each short segment represents an interpretation of ~75–100 along-track signal photons (Jasinski et al., 2023). This results in segment lengths varying, from approximately 30 m to a few hundred m, depending on the terrain and beam strength. However, this variability is consistently smaller than GMAI spatial resolution, which results in overgeneralization of the lake surface. Strong beams yield smaller short segment lengths because they have a higher energy when compared to weak beams, where strong beams are about four times stronger than the weak beams (Neumann et al., 2019). Therefore, our algorithm focuses on classifying strong beam data points from the ATL13 product, omitting the weak beam data to enhance the accuracy and reliability of the ice-water discrimination algorithm. ATL13 data was obtained from the National Snow and Ice Data Center (NSIDC) for four years of data covering the Great Lakes region (01/01/2019–04/30/2019; 01/01/2020–04/30/2020; 01/01/2021–04/30/2021; 01/01/2022–04/30/2022; 01/01/2023–04/30/2023) (Jasinski et al., 2023). We performed further spatial filtering on the ATL13 data to obtain Great Lake specific data because ATL13 includes data from all bodies of water in the surrounding area, like additional lakes, rivers and streams. The data analyzed was dependent on both the spatial bounds of the Great Lakes and coincident data from ICECON and Landsat 8/9. A total of 2,608,446 ICESat-2 short segment points were used in this study, as shown in Figure 2 (consisting of 708 individual strong beam tracks or 236 ICESat-2 passes).

The surface photon rate parameter used in the ICESat-2 sea ice algorithms is not directly available in ATL13, so a variation of the surface photon rate is used, which is designated the signal photon rate in this study (Koo et al., 2023; Kwok et al., 2022). The surface photon rate, used in the sea ice algorithms, is the number of signal photons per pulse (Kwok et al., 2021). The signal photon rate, used in this algorithm, is a measure of the number of signal photons per meter for each short segment. The primary difference between the photon rate used in the sea ice algorithms and this algorithm is the photon rate used in the sea ice algorithms is relative to the number of photons in the shot (Kwok et al., 2021). The photon rate used in this algorithm is relative to the short segment length (Jasinski et al., 2023). The rate of received signal photons, whether it be relative to the number of photons in the shot or the short segment length, is indicative of the apparent surface reflectance of a short segment wherein the larger the rate the higher the surface reflectance (Kwok et al., 2021). For the signal photon rate, used in this algorithm, the higher the signal photon rate is, which is a result of smaller short segment lengths, the higher the apparent surface reflectance. Observing a photon rate relative to the length of the short segment allows for more representative observation of brightness because the ATL13 short segments are derived from averaged ~75–100 signal photons and have varying lengths (Jasinski et al., 2023). This makes the signal photon rate predominantly dependent on the length of the short segment (Jasinski et al., 2023).

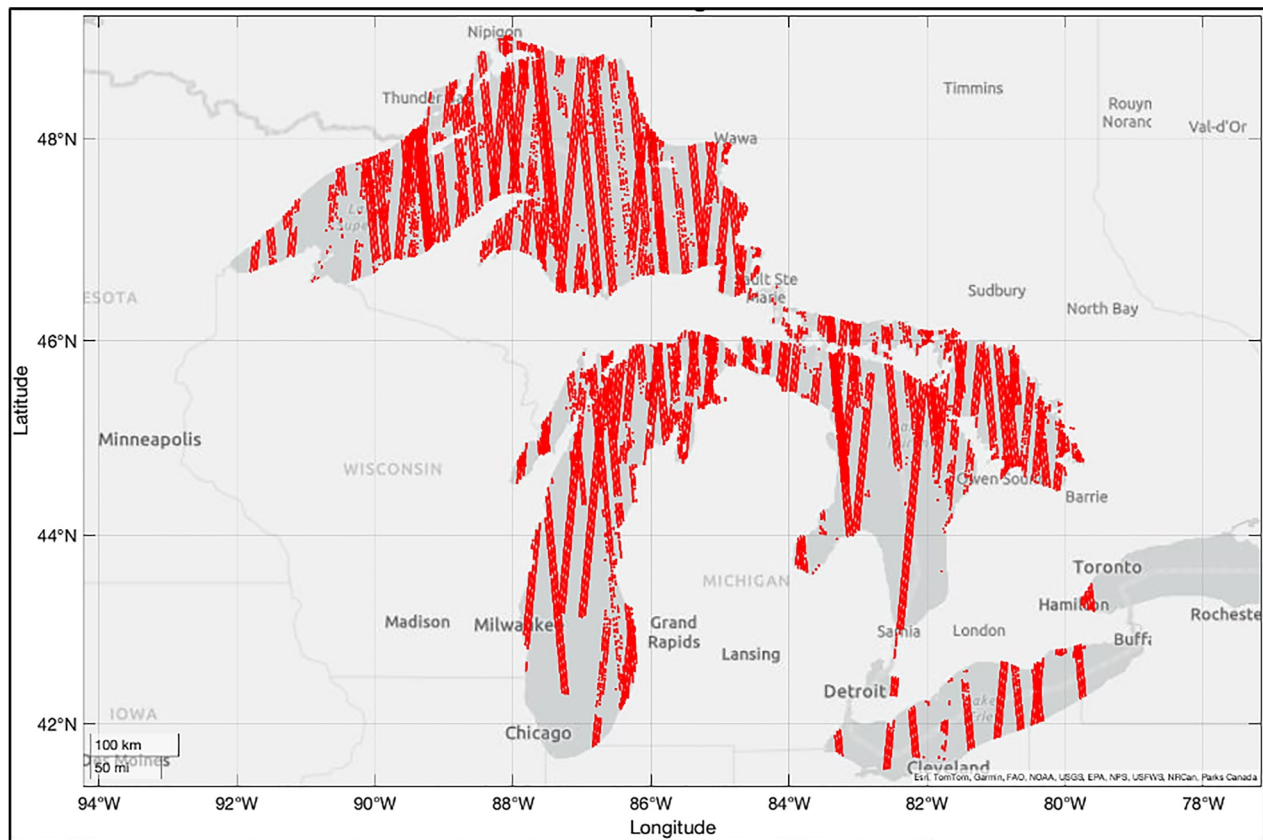


Figure 2. Map of all ICESat-2 ATL13 strong beam data used in this study.

The standard deviation of the height distribution for each short segment from ATL13 is used as well. In the data-driven machine learning and decision tree based sea ice algorithms, the width of the Gaussian height distribution is used (Koo et al., 2023; Kwok et al., 2022), but again this parameter is not directly available from ATL13, though a variation of it can be used. The standard deviation of the height distribution is indicative of the surface roughness which would capture the presence or absence of waves and potentially the extent of ice coverage. Higher standard deviations correspond to rougher surfaces, like large wave heights or in rarer cases thick ice, and smaller standard deviations are indicative of smoother surfaces like ice or a smooth water surface (Koo et al., 2023).

The background photon rate is also not provided by ATL13, but the background rate flag is available. This flag indicates an interval of background photons observed per bin per short segment. The background rate flag is a measure of the non-signal identified photons and can be used to describe the likelihood of ice for each short segment wherein as the background photon counts per bin per short segment increases, the likelihood of ice increases (Jasinski et al., 2023). This variable must be converted to a categorical value, as opposed to a numerical value, when integrating it into the neural network.

The segment temperature is a variable available in ATL13 which is indicative of the state of the water surface. This variable is derived from the ATL09 surface (skin) temperature using a nearest neighbor linear interpolation from the GEOS-5 model (Jasinski et al., 2023; Palm et al., 2018). The segment saturation fraction is obtained from ATL13 and is a measure of the fraction of geosegments (~20 m) within a short segment that are fully saturated (≥ 16 photons/shot) which means the return exceeds the expected number of photons. A larger number of saturated geosegments is an indication of higher returns which can be interpreted as a result of more reflective surfaces like ice.

3.2. ICECON

ICECON data obtained through the NOAA Great Lakes Environmental Research Laboratory (GLERL) THREDDS Data Catalog was used to classify surfaces to train the model (Leshkevich & Nghiem, 2013). ICECON is primarily used by the US Coast Guard for ice cutting operations and ship transit assistance in the Great Lakes region. The ICECON product is the primary source of ice classification for the Great Lakes and is maintained and produced by the NOAA GLERL and National Environmental Satellite, Data, and Information Service. This product uses SAR satellite imagery (RADARSAT and Sentinel 1A/1B) to classify different types of ice when there are passes over the Great Lakes. Because of this limited geographic coverage, the use of this product only extends to testing and validation in this region and cannot be used for global and continuous observations of lake ice. The primary advantages of using the gridded ICECON product include not only its accuracy and resolution but also its ability to capture the presence of clear, newly formed, thin ice on the lake. One major drawback to using ICECON for validation and development of an ICESat-2 based product is that the temporal resolutions of each satellite differ, meaning only ICECON images overlapping in space and time with ICESat-2 can be applied for this study. To achieve this, we collected ICECON data from January through April of 2019–2023. We used spatial and temporal (within 24 hr) filtering functions to acquire ICECON data that corresponded to the ICESat-2 data. Corresponding ICECON classifications were then mapped to each ICESat-2 short segment point using a polynomial function that returns a grid location from given latitude and longitude points (see Data Availability). ATL13 data that did not have a coincident ICECON image were removed. We did not extrapolate any ICECON surface classifications. Raw ICECON returns the categories of free water (no ice present), new lake ice, pancake ice, consolidated flows, lake ice with snow, and brash rather than a binary classification. All ATL13 data was labeled as either “ice,” which encompasses all ice types, or “free water.” Through these methods, we created a data set containing all possible ICECON classified ICESat-2 points between 2019 and 2023. We used a total of 538 ICECON images for training and testing purposes, covering the 236 ICESat-2 passes shown in Figure 2.

ICECON surface classifications were used as the true surface classification, as described by Table 1, in the predictive NN and are used as the primary classification method in the overall algorithm, described in Figure 3 and Text S1 in Supporting Information S1. Additionally, we used ICECON to assess the performance of the current classification system in place for ATL13, GMAI.

3.3. Landsat 8/9 OLI-TIRS Collection 2 Level 2 Data

The algorithm presented here uses variables from Landsat 8/9 OLI-TIRS Collection 2 Level 2 Data as features, or inputs, to one of the ice-water discrimination NN described in Section 4.2. This data was collected from the U.S. Geological Survey. Landsat 8/9 both carry the Operational Land Imager with nine spectral bands and the Thermal Infrared Sensor (TIRS for Landsat-8 and TIRS-2 for Landsat-9). The spatial resolution of the bands used in this study are 100 m, which are then resampled in the collection 2 data set to 30 m, which is on the same scale as ATL13 data, and has a combined revisit time of 8 days (16 days for each satellite), which allows for monitoring global changes. We obtained Landsat 8/9 data for all ICESat-2 passes that aligned spatially and temporally (within 16 days prior to the launch of Landsat 9, and within 8 days after) with the ICESat-2 data used in this study. A total of 200 Landsat scenes were used in the training of the neural network.

Blue (0.450–0.51 μm), green (0.53–0.59 μm) and red (0.64–0.67 μm) channels from Landsat 8/9 are used in one of the NN in this algorithm as shown in Table 1. Ice often appears very bright in all channels of visible imagery, relative to water surfaces which exhibit lower values in these channels due to higher absorption of light (Katlein et al., 2015). Data with lower (darker) values for these channels are an indication of open water, rather than ice. Landsat 8/9 thermal surface radiance (TRAD; B10 (10.6–11.19 μm) and B11 (11.5–12.51 μm)) is used (in addition to the surface temperature) because thermal radiance is a function of the emissivity of the object and the actual temperature. Over freshwater lakes the ice and the surface of the water can be very close in temperature, but the emissivity of these surfaces can vary (Kang et al., 2014). The thermal radiance of a non-blackbody is dependent on the emissivity and the temperature, as defined by Stefan's Law. The thermal surface radiance recorded by the L2 product in Landsat 8/9 has units of radiance that is integrated over the solid angle (steradians) and the wavelength ($\text{W}/\text{m}^2/\text{sr}/\mu\text{m}$), but the relationship would remain the same (Teillet, 2024). Although the allowed time difference for overlapping Landsat 8/9 data may be large, larger time differences are accounted for in the ice-water discrimination neural network. The time difference between the coincident Landsat 8/9 and

Table 1
Summary of Algorithm Inputs

Source	Variable	Description
ICESat-2 ATL13	Signal Photon Rate	Measured in signal photons per meter for each short segment; indicative of surface reflectance or brightness; variation used in sea ice algorithms (ATL07/10/20/21)
	Short Segment Standard Deviation	Measured in meters using a deconvolution algorithm; indicative of surface roughness; sea ice algorithm (ATL07/10/20/21) used width of the height distribution from convolution approach
	Background Rate Flag	Indicative of shadows, specular reflections or atmospheric corruptions and likelihood of ice; original sea ice algorithm (ATL07/10/20/21) used background photon rate
	Segment Temperature	Indicative of ice present, interpolated from ATL09 MET surface (skin) temperature
	Segment Full Saturation Fraction	Fraction of fully saturated geosegments within a short segment; indicative of specular reflective surfaces
	Surface Classification (GMA5I)	
Landsat 8/9	B2 (blue channel; 0.450–0.51 μm)	Brighter values are an indication of ice, darker values are an indication of open water
	B3 (green channel; 0.53–0.59 μm)	Brighter values are an indication of ice, darker values are an indication of open water
	B4 (red channel; 0.64–0.67 μm)	Brighter values are an indication of ice, darker values are an indication of open water
	TRAD (Thermal Surface Radiance; 10.6–11.19 μm and 11.5–12.51 μm)	Indication of surface type, relates temperature and emissivity
ICESat-2 ATL13 and Landsat 8/9	Landsat 8/9 ICESat-2 Time Difference	Accounts for large time differences between the closest corresponding Landsat 8/9 footprint
ICECON	Ice Type	Used for labeling training, testing and validation data

Note. Green cells indicate parameters used in one or both neural networks. Yellow cells correspond to the current classification method of ATL13 data. Cyan cells are used for developing neural networks but are not an input parameter.

ICESat-2 passes is an input parameter to the neural network as well. This parameter should aid in the NN learning process, where smaller weights would be placed on Landsat 8/9 features when making predictions if the time difference is large. Because Landsat 8/9 data is stored on a grid, each ATL13 data point was projected to the mapping coordinate system of the coincident Landsat 8/9 scene to obtain the corresponding input parameters.

3.4. Overview of Parameters

In developing the algorithm, key variables from ATL13, Landsat 8/9, and ICECON were obtained for the ICESat-2 passes used (Table 1). The classification algorithm incorporates these variables to distinguish between ice-covered and open water surfaces. Some features of the two NN, described in Section 4.2, are consistent with the features used in both sea ice algorithms mentioned above, and some are additional parameters that were added to enhance the predictive performance for large lakes.

4. Methods

We developed three methods of classification that are integrated into the overarching algorithm (Figure 3). The primary method of classification for ATL13 data over the Great Lakes is using ICECON, which is the closest representation of the true surface classification currently available for this region. Two NN were trained for the surface classification algorithm; (a) one using both ICESat-2 and Landsat 8/9 data for optimal determination (i.e., “IL”) in Section 4.2.1, and (b) one using ICESat-2 data only (i.e., “I”) in Section 4.2.2 (Figure 4). We observed the importance of the neural network input parameters using feature permutations (Section 4.1). We then evaluated the performance of the two NN using a series of performance metrics relative to ICECON surface classifications.

4.1. Feature Investigation

ICESat-2's ATL13 inland water surface product provides several parameters that are indicative of surface characteristics. An investigation into the feature selection allowed for the identification of potential correlations between input parameters for the algorithm. It also ensured that each input parameter was contributing to the accuracy of the neural network in a positive manner. We used feature permutations to determine if a particular

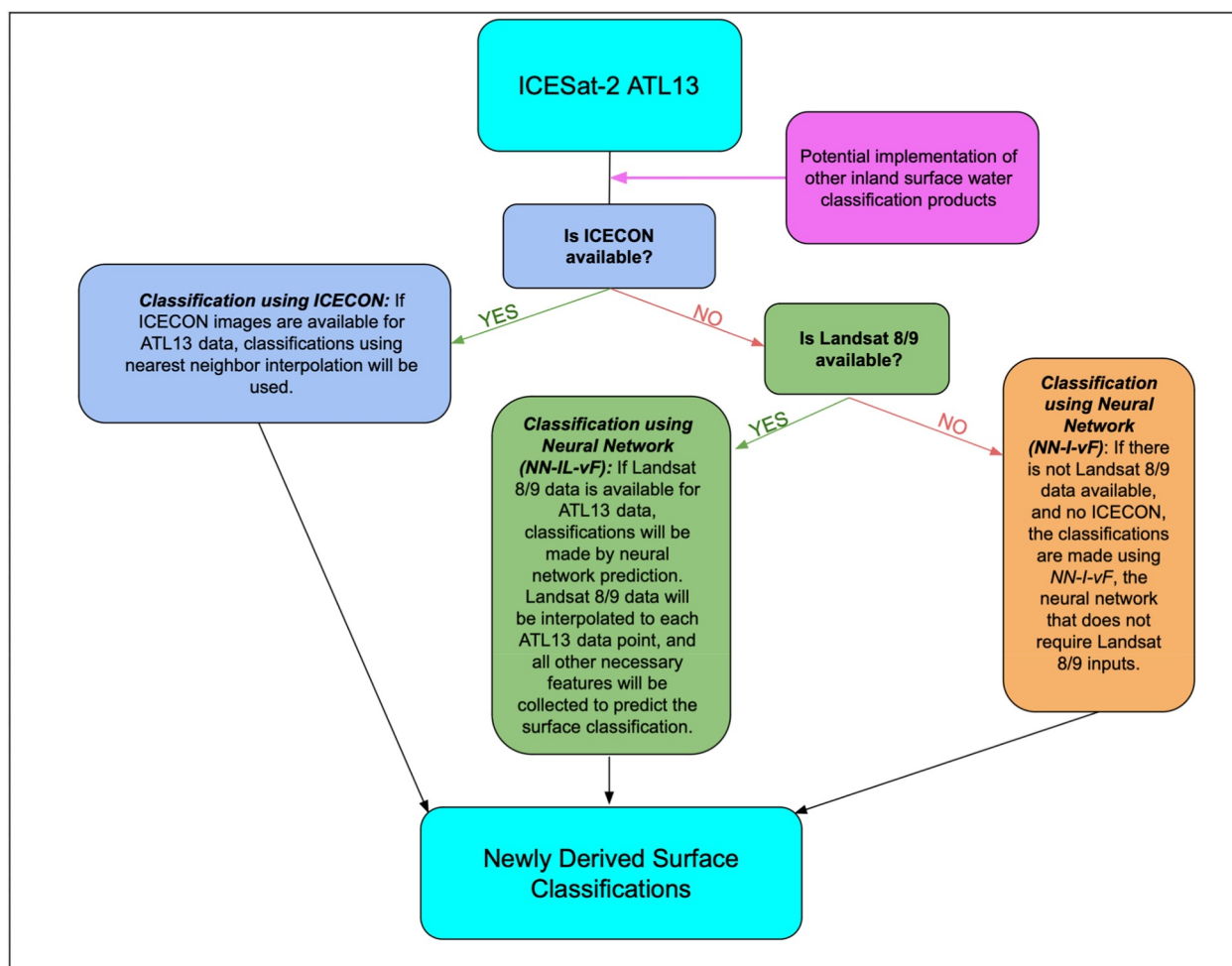


Figure 3. Overview of the surface classification algorithm.

input parameter in a neural network contributes positively or negatively to the predictive accuracy of the model (Molnar, 2020). In this feature permutation investigation, we isolated a random subset of data (20% of all available data), for testing purposes, and used the remainder of the data for training. The values of one input parameter, or feature, in the training data set are randomly shuffled and the rest of the input parameters remain the same, in their original order. This new training data set can be known as permutation_i. We trained a neural network using the permutation_i data set and then the accuracy of the network was tested. If the accuracy of the permutation_i trained network decreases relative to the unshuffled prediction accuracy, the feature that was shuffled was considered to increase the predictive accuracy of the model and was then deemed a viable feature. If the shuffled model accuracy increases or remains the same, the feature does not contribute to the predictive accuracy of the unshuffled model and can be deemed necessary. This process is then repeated for all input parameters in the model, which resulted in 10 permutations.

4.2. Training the Neural Networks

Prior to training the NN, the ICESat-2 and coincident data must be in the proper format and a number of pre-processing steps are required. The formatting and preprocessing steps are described in Section 3.

4.2.1. Training NN-IL-v0/F

We acquired coincident Landsat 8/9 data (blue, green, red, and TRAD channels) to develop a neural network using ICESat-2 and Landsat 8/9 data. Two NN were trained using ICESat-2 and Landsat 8/9 data. The first neural network, *NN-IL-v0*, or the initial version of the model, was trained before producing the final version of the neural

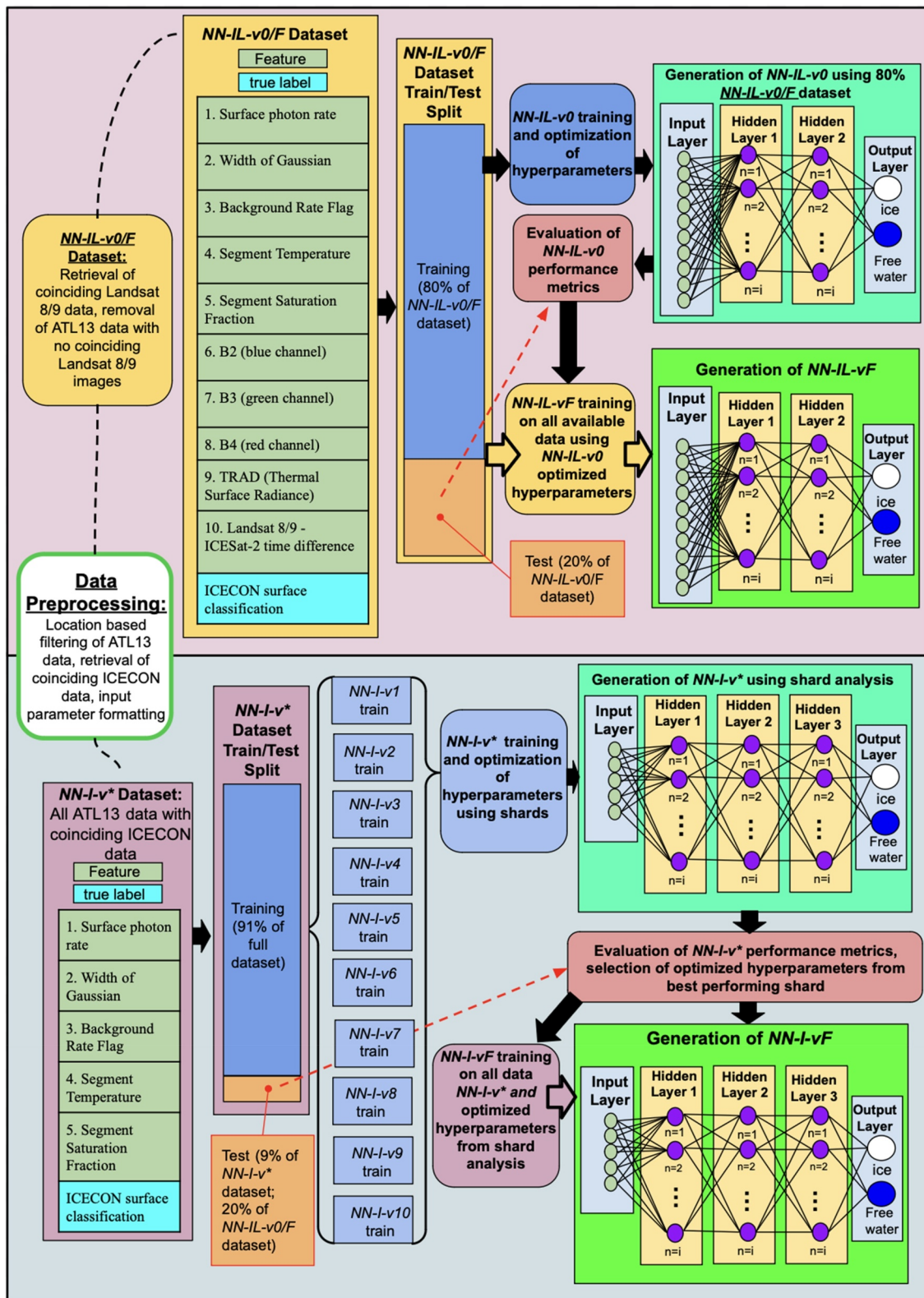


Figure 4. Flowchart of neural network training process.

network (*NN-IL-vF*), to observe testing metrics and viability of the method. We used a random subset of 80% (427,920 points) of the full data set (*NN-IL-v0/F Data set*) to train *NN-IL-v0*, and 20% of the data set (106,980 points) was reserved to evaluate the performance of *NN-IL-v0* after it was trained (see Table S1 in Supporting Information S1 for further details). No testing data was used in the training of *NN-IL-v0*. Training the *NN-IL-v0* on a subset of the *NN-IL-v0/F Data set* allows for performance metrics to be assessed, which will give a strong indication of the performance *NN-IL-vF*, which is trained on the entirety of the *NN-IL-v0/F Data set*. The features used as input parameters are listed (1–10) in the *NN-IL-v0/F Data set* region of Figure 4 and Table 1. The ICECON classifications (bottom of the *NN-IL-v0/F Data set* region), were considered to be the true labels for the surface classification.

A crucial component of training a reliable neural network is tuning the hyperparameters (Joy et al., 2016). In this study, we used hyperparameter optimization to determine the most suitable hyperparameters for the data at hand. Hyperparameter optimization methods, such as grid search, random search, and Bayesian optimization, help in systematically exploring the hyperparameter space to find the best configurations for model performance (Yang & Shami, 2020). Here we used Bayesian optimization (Szandala, 2021). The hyperparameter optimization used in this study allows for the neural network to vary the number of fully connected layers, the activation function, the regularization strength, and whether or not the data is normalized in order to find the optimal configuration for the produced neural network (Joy et al., 2016). The hyperparameter optimization functions in the MATLAB Deep Learning Toolbox tests for a maximum of three hidden layers, which is acceptable for this data set because each of the features give some indication of the surface and therefore will not exhibit the data complexity that typically requires a larger number of hidden layers. The optimized neural network configuration should balance model complexity and computational efficiency, ensuring the model can capture the underlying patterns without overfitting (Szandala, 2021). Furthermore, the optimized layer sizes were chosen to represent the complexity of the data, as too few neurons might lead to underfitting, whereas too many could result in overfitting and increased computational costs (Zhang et al., 2019). This setup, supported by regularization techniques and proper initialization, enhances the model's ability to generalize from the training data to unseen data (Szandala, 2021).

We trained the *NN-IL-v0* using hyperparameter optimization and then evaluated using the testing data (20% remaining) to ensure performance metrics were sufficient for reliable surface classification. The performance metrics consist of accuracy (Equation 1), precision (Equation 2), recall (Equation 3), F1 score (Equation 4) and the intersection over union (IoU, Equation 5), all of which are relative to the ICECON classification. Accuracy gives a measure of the overall correctness of the model. Precision measures the accuracy of the positive, or ice classified, predictions. Recall measures the ability of a model to identify relevant instances. The F1 score is the harmonic mean of precision and recall. Finally, the IoU allows for an evaluation of overlap ad segmentation accuracy. Considering all these metrics in unison allows for a more comprehensive understanding of the model's performance. Performance metrics for the GMAI classification, currently employed in ATL13, and the neural network classification developed in this study were calculated and compared. A prediction was considered to be true positive if the ice was correctly identified as ice and a false positive prediction corresponds to when ice was incorrectly identified as free water. A true negative prediction corresponds to free water data that was correctly identified as free water and a false negative corresponds to when free water was incorrectly identified as ice.

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total Number of Samples}} \quad (1)$$

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (2)$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (3)$$

$$\text{F1 Score} = 2 \left(\frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \right) \quad (4)$$

$$\text{IoU} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives} + \text{False Positives}} \quad (5)$$

Two types of cross validation experiments were performed on the same training data set used for *NN-IL-v0* to further support the results of this study (see Text S2 and Figure S1 in Supporting Information S1 for further details). A spatial cross validation analysis was performed, where we used a leave-one-lake out approach. This resulted in training five new NN for each lake left out, which we could then test on both the left-out-lake data (validation data) and the testing data. We also performed temporal cross validation experiments, where we used a leave-one-year out approach, which resulted in five additional networks that could then be tested on the corresponding validation and testing data.

Once we trained a model that sufficiently outperformed GMASI, a final neural network (*NN-IL-vF*) was trained using the optimized hyperparameters from *NN-IL-v0* with all the available data in the *NN-IL-v0/F Data set*. Retraining the neural network allows for all available data to be considered when creating weights and biases for each neuron in the network. By training the *NN-IL-vF* on the entirety of the *NN-IL-v0/F Data set*, the network will be exposed to a broader set of data, which should increase the generalizability of the model and enhance its performance. Metrics observed in the performance of the initial neural network will give a strong indication of how the final neural network performs.

Using all available ATL13 data with coincident ICECON and Landsat 8/9 data does not allow for performance metrics to be evaluated because no data is preserved for testing in the training of *NN-IL-vF*. In order to ensure the final neural network (*NN-IL-vF*) is comparable to the initial neural network (*NN-IL-v0*), a direct comparison of the two model classifications was made. We made predictions using the two models on a broader data set, that consisted of the data both models were trained on, as well as ATL13 data that had coincident Landsat 8/9 data but not coincident ICECON data. This means a fraction of the data set was used to train the networks and the remaining fraction of the data set has not been seen by one or both of the networks because there were no overlapping ICECON data available for these points (769,509 total points used for comparison, 534,900 points [69.5% of total] were used in training *NN-IL-vF*, and 427,920 points [55.6% of total] were used in training *NN-IL-v0*).

4.2.2. Training NN-I-vF

We trained a second set of NN to predict ICECON classified surfaces using exclusively ICESat-2 ATL13 data, which is denoted as *NN-I-vF*. *NN-I-vF* was ultimately trained on the entire *NN-I-v* Data set*, which consisted of all data covering the Great Lakes from the 2019–2023 ice seasons that have corresponding ICECON images. Because overlapping Landsat 8/9 data is not necessary for the training of this neural network, the training and testing data sets used are much larger than those mentioned in Section 4.2.1 to train *NN-IL-v0*. However, we extracted the same testing data set used to evaluate the performance of *NN-IL-v0* from the *NN-I-v* Data set* (containing 1,180,359 points) for the purpose of having directly comparable performance metrics. This extracted testing data set made up ~9% of the *NN-I-v* Data set*, leaving ~91% of the data to be used for training.

Because this training data set was so much larger than the one used in the training of *NN-IL-v0*, the computational cost for optimizing hyperparameters increased significantly. To address this increased training time, we used a shard analysis to optimize hyperparameters for the training of *NN-I-vF* (Singh et al., 2024). This shard analysis consisted of partitioning the training data into 10 random shards, where then each shard was used to train a neural network using hyperparameter optimization (*NN-I-v(1-10)*) to predict ICECON surface classifications. The shard based NN were then evaluated using the preserved testing data set (see Figures S2–S12 in Supporting Information S1 for performance metrics). The hyperparameters for the best performing shard were used to train the final neural network, *NN-I-vF*. Spatial and temporal cross validation experiments, similar to those described in Section 4.2.1, were performed on the training data set for *NN-I-vF* to ensure the generalizability of *NN-I-vF* (see Text S2 and Figure S1 in Supporting Information S1). We compared the performance metrics from the *NN-I-vF* to those observed in *NN-IL-v0*. We calculated performance metrics for both *NN-IL-v0* and *NN-I-vF* to assess temporal performance anomalies for the entire testing data set and for individual years, wherein the testing data corresponding to each year was isolated and evaluated for each method, to identify any year-to-year variation of biases. *NN-IL-vF* and *NN-I-vF* were then integrated into the overall algorithm where surface classifications are made based on what coincident data is available (see Figure 3 and Text S1 in Supporting Information S1 for further details of this algorithm as well as Figure S13 in Supporting Information S1 for computational resources and time required for each neural network).

Table 2
Feature Permutation Accuracy

Feature permutation	Test accuracy (%)
<i>Unshuffled Features</i>	94.221
Width of the Gaussian	94.144
B2	94.127
B4	94.007
Background Photon Rate Flag	93.966
B3	93.96
Saturation Fraction	93.953
Surface Photon Rate	93.816
Landsat 8/9 Time Difference	93.134
TRAD	93.002
ATL09 Segment Temperature	92.964

Note. The unshuffled features row (italicized) corresponds to the control, where no permutations were performed. Accuracy of the control in cyan, highest tested permutation accuracy in light orange, lowest permutation accuracy in dark orange.

5. Results

5.1. Feature Investigation Results

Investigation of the input parameters was carried out using a feature permutation approach (Table 2). While all permutations resulted in a decrease in accuracy, indicating their significance to the algorithm, the models trained on shuffled *ATL09 Segment Temperature*, *Thermal Surface Radiance*, and *Landsat 8/9 Time Difference* exhibit the highest decrease of accuracy. This is an indication that these features are most influential in making accurate predictions. It is important to note that the *Landsat 8/9 Time Difference* was one of the most influential parameters seen in this feature permutation investigation. It is expected that the *Landsat 8/9 Time Difference* influences all Landsat 8/9 parameters because of the relationship between this time difference and how representative the Landsat 8/9 data is of the surface ATL13 data represents. In other words, as this time difference increases, Landsat 8/9 parameters become less representative of the surface that ATL13 data is describing, and should result in these features holding less weight in the surface prediction. The average time difference between each ICESat-2 data point and its coincident Landsat 8/9 data for all data used in this study was ~5.5 days, but is smaller for the years 2022–2023 because of the launch of Landsat 9 in September 2021 (~5.4 days) and larger for years 2019–2021 (~6.1 days).

5.2. Model Performance Metric Evaluation

Confusion matrices of *NN-IL-v0*, *NN-I-vF*, and GMASI, from which accuracy, precision, recall, F1-score, and IoU are derived, provide analysis and comparison of the three classification methods (Figure 5). For both the *NN-IL-v0* and *NN-I-vF*, the difference between the misclassification of ice and the misclassification of free water is small, indicating there is no bias toward misclassifying a particular surface. The IoU of the *NN-IL-v0* (88.2%) was observed to be the highest followed by the IoU of *NN-I-vF* (79.8%), indicating good agreement between the predicted and true classes relative to the IoU of GMASI (51.7%). The confusion matrix for GMASI shows substantial bias toward misclassifying ice as free water, where a large proportion of true ice surfaces were classified as free water (17.2%) and a smaller amount of misclassification can be attributed to free water classified as ice (5.6%).

We used testing data (consisting of 20% of the *NN-IL-v0/F Data set* or 9% of the *NN-I-v* Data set*) exclusively for calculating the performance metrics of *NN-IL-v0* and *NN-I-vF*, meaning the NN have not been exposed to this set of data. A detailed summary of these performance metrics is presented in Table 3. We also evaluated the GMASI performance in this table, so that comparisons could be made between the three classification methods. Cross validation performance metrics are presented in Tables S2–S5 in Supporting Information S1.

With the exception of precision, the *NN-IL-v0* outperformed *NN-I-vF* and GMASI for all individual years. *NN-I-vF* has the second-best performance when considering these metrics, and GMASI has the lowest performance, for all individual years. When observing the precision, for the data tested in the 2021 subset, GMASI labeled all data points as ice and received a 100% precision as a result (False Positive = 0%, False Negative = 56.31%). This is an indication that GMASI is under-predicting ice coverage, because a large proportion of the misclassifications are a result of false negative predictions (model classification of free water, true classification is ice). Similar to the case of the precision for 2021, in the data from 2020 to 2022, there is a comparatively larger false negative percent relative to the false positive percent which would result in a higher precision value relative to the NN precision (False Positive 2020 = 1.94%, False Negative 2020 = 13.38%; False Positive 2022 = 5.7%, False Negative 2022 = 36.75%). Again, this is an indication that GMASI is under-predicting ice coverage and overgeneralizing fractional ice-covered surfaces.

The *NN-IL-v0* outperforms both the *NN-I-vF* and GMASI for every metric calculated, when evaluating the performance of the NN on the entire testing data (2019–2023). The *NN-I-vF* outperformed GMASI in every metric except precision, which is likely a result of GMASI under-predicting ice, as mentioned earlier. The accuracy serves as an evaluation of the overall correctness of the model, and the overall accuracy of *NN-IL-v0* is

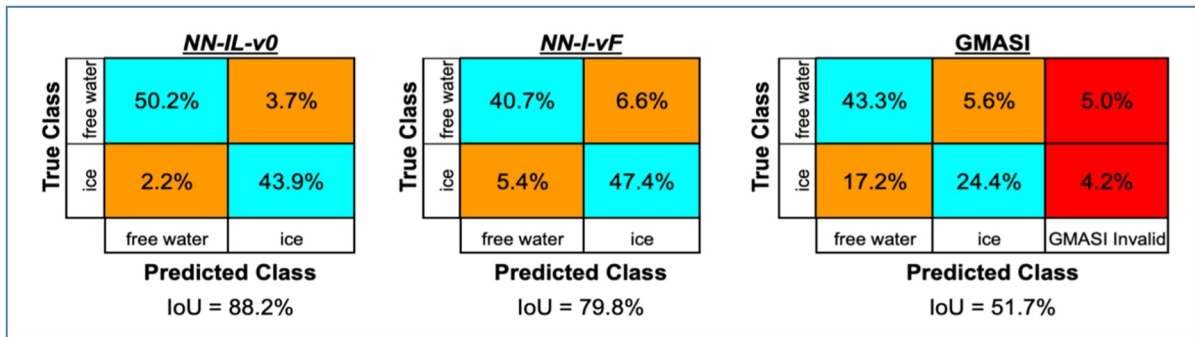


Figure 5. Confusion matrices of NN-IL-v0 (left), NN-I-vF (middle) and GMASI (right).

94.1%, which is higher than the accuracy of GMASI by 26.46%. The overall accuracy of the *NN-I-vF* is 88.01% which is higher than the overall accuracy of GMASI by 20.37%. In addition, GMASI classifies 9.26% of the data as invalid due to either the coarse resolution or insufficient model inputs, whereas the NN do not lose any data for the Great Lakes because of the lake bounds used in the algorithm.

We also analyzed which ice types were misclassified the most by *NN-IL-v0* and *NN-I-vF* using the error rates normalized by the number of samples belonging to an ice type (see Figure S14 in Supporting Information S1). For both NN the highest normalized error rates were observed for new lake ice, followed by brash ice, consolidated flows, and pancake ice.

The performance of *NN-IL-v0* and *NN-I-vF* was much better than the current classification method in place (GMASI), such that it was determined the final neural network (*NN-IL-vF*) could be trained using all the available data in the *NN-IL-v0/F Data set*. To ensure the *NN-IL-v0* and *NN-IL-vF* were performing in a similar manner, a

Table 3
Yearly and Cumulative Performance Metrics of NNv0 and GMASI

Test year(s)	Prediction method	Accuracy %	Precision %	Recall %	F1-score	GMASI invalid %
2019	<i>NN-IL-v0</i>	94.66	95.90	97.25	0.9657	N/A
	<i>NN-I-vF</i>	86.24	88.07	94.00	0.9094	N/A
	GMASI	63.61	67.30	75.26	0.7527	2.74
2020	<i>NN-IL-v0</i>	95.51	96.87	97.16	0.9701	N/A
	<i>NN-I-vF</i>	90.41	93.00	94.20	0.9360	N/A
	GMASI	77.01	97.40	84.43	0.9045	9.06
2021	<i>NN-IL-v0</i>	99.16	99.28	98.10	0.9868	N/A
	<i>NN-I-vF</i>	85.15	98.80	64.45	0.8087	N/A
	GMASI	43.49	100	35.90	0.5284	0.46
2022	<i>NN-IL-v0</i>	89.40	70.51	81.69	0.7569	N/A
	<i>NN-I-vF</i>	82.11	53.68	64.08	0.5842	N/A
	GMASI	52.69	75.95	32.88	0.4589	8.43
2023	<i>NN-IL-v0</i>	96.95	97.21	98.32	0.9776	N/A
	<i>NN-I-vF</i>	92.03	94.79	93.67	0.9423	N/A
	GMASI	77.58	93.28	88.66	0.9091	11.01
2019–2023	<i>NN-IL-v0</i>	94.10	93.08	95.85	0.9445	N/A
	<i>NN-I-vF</i>	88.01	87.81	89.73	0.8876	N/A
	GMASI	67.64	88.53	71.21	0.7893	9.26

Note. Blue, orange, and red cells correspond to the best, middle, and worst performing surface classification method for each year and metric. Summary performance metrics for the four ice seasons used in this study are provided in bold.

direct comparison of the two model classifications was made for ATL13 data from 2019 to 2023 over the Great Lakes ice formation seasons. When making predictions on the full data set 88% of predictions were the same.

5.3. A Comparison of Surface Classification Methods in Lake Superior

In the previous section, we assessed the overall test performance for the neural network relative to GMASI and observed results that could indicate the neural network is a better alternative for surface classification as opposed to GMASI. As a representative example, Figure 6 shows a comparison of the methods considered in this study. The data plotted in Figure 6 show ATL13 data from an ICESat-2 pass on 02/13/2023 over the Eastern side of Lake Superior. Among the four methods, *NN-IL-vF* most accurately represents the true surface classification (ICECON), wherein it captures the coastal ice that ICECON shows (around 47°28'N and 46°45'N), and the free water in between, achieving an accuracy of 92% for this subset of data. The *NN-I-vF* classification shows very similar results but captures some fractional ice patterns not exhibited by ICECON in the coastal regions and achieved an accuracy of 84.7%. It also fails to pick up the ice of the southern coast near 46°45'N. Surface classifications made by both NN show some indication of fractional ice coverage in between 47°15'N and 47°51'N. The surface classifications made by GMASI, with an accuracy of 69.3%, show no ice cover between 47°28'N and 46°45'N, which does not follow the true surface classifications from ICECON. There is also a fraction of data that would be dropped in analysis due to GMASI invalid classifications.

6. Discussion

We explored an alternative surface classification algorithm for ICESat-2's ATL13 short segment, strong beam data in this investigation. By introducing pre-existing surface classification data products, external sources of data, and data driven machine learning, a surface classification algorithm that outperforms the current surface classification method for ATL13 (i.e., GMASI) was developed and applied to the Great Lakes. Performance metrics for *NN-IL-v0* and *NN-I-vF* were calculated using identical testing data sets, to ensure direct comparisons of the two networks' performance. When directly comparing the test statistics, the *NN-IL-v0* outperformed *NN-I-vF* and GMASI by 6.09% and 26.46%, respectively. *NN-I-vF* outperformed GMASI by 20.37%. Our analysis showed that using Landsat 8/9 optical and thermal imagery when casting predictions improves surface classification accuracy by over 26%, but even without the use of Landsat 8/9, surface classification accuracy improves by over 20% relative to the current surface classification method in place. This increased accuracy suggests that using the networks trained in this study could serve as an effective alternative to GMASI (Romanov, 2017). Other than its relatively low accuracy, GMASI has a tendency to under-predict ice coverage (Figures 4 and 6). This bias toward predicting free water when the true surface is ice could have cascading effects if ICESat-2 ATL13 data was being used operationally. Because large lakes, particularly the Great Lakes, are used for both commercial shipping and recreation, misclassifications of ice as free water could impact the safety of those utilizing these lakes (Anderson et al., 2018; Yeo et al., 2024). Additionally, almost 10% of ICESat-2 ATL13 data is lost as a result invalid classifications made by GMASI, because of its coarse resolution and predictive ability. Using NN that use higher resolution data, or data products like ICECON, to make surface classifications in place of GMASI could not only improve the surface classification accuracy but also minimize the amount of data discarded. Computation demands for both training and making predictions with these NN are minimal as well (Figure S13 in Supporting Information S1). This minimal computational requirement enables the opportunity to retrain each model as new ice season data becomes available and provides an efficient alternative for surface classification of ICESat-2 ATL13 data.

When developing these NN, we used parameters similar to those used in the two sea surface classification algorithms (Koo et al., 2023; Kwok et al., 2022). To ensure the inputs of our algorithm were providing beneficial information for surface classification, feature permutations were used to understand the contribution of each input parameter. The results showed ATL09 surface temperature, TRAD, and the Landsat 8/9 to ICESat-2 time difference contributed the most to the predictive accuracy of the networks, but all permutation iterations decreased the predictive accuracy to some degree, meaning each parameter has a positive contribution to the networks. The accuracy of the permutations of ICESat-2 derived input parameters decreased relative to the unshuffled iteration, which aligns with the expectations and observations of the sea ice algorithms (Koo et al., 2023; Kwok et al., 2022).

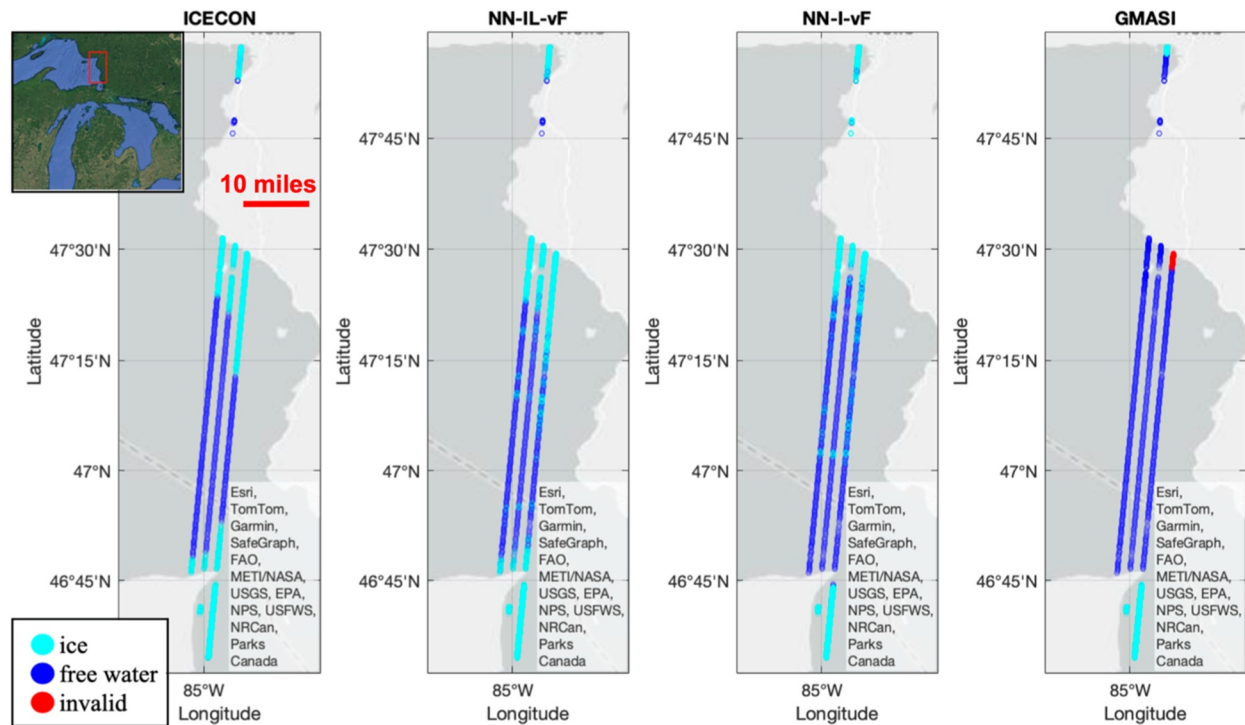


Figure 6. Surface classification method comparison ICESat-2 tracks from 02/13/2023 on Eastern Lake Superior.

The neural networks *NN-IL-v0/F* and *NN-I-vF* were trained using a wide variety of data that spanned 4 years of ice season ICESat-2 data to minimize the biases in the training process. These NN were also spatially and temporally cross validated to further show generalizability. The full data set consisted of 2,608,446 ICESat-2 short segment points, from 708 individual strong beam tracks, which is about eight times the amount of data used in sea ice machine learning algorithms for surface classification (Koo et al., 2023). When training *NN-IL-vF*, we chose to maintain the optimized hyperparameters from *NN-IL-v0* to allow for training on the maximum amount of data while developing the neural network with architecture similar to *NN-IL-v0*. We assumed that the performance of *NN-IL-v0* should be consistent with the performance of *NN-IL-vF*, because the *NN-IL-vF* is trained on the full data set, where 80% of the data used to train *NN-IL-vF* would be the same as its initial version. To show how similar the *NN-IL-v0* and *NN-IL-vF* were, we made predictions using each network on a broader data set of overlapping Landsat 8/9 and ICESat-2 data. 88% of the predictions between the two networks were the same, which is an indication that the two networks are performing and learning similar patterns to make predictions. The 12% of data that were not classified the same could be due to the difference in the amount of data each network was exposed to. The *NN-IL-v0* was only exposed to 80% of the data set that the *NN-IL-vF* was trained on, and the increased amount of data used to train *NN-IL-vF* should decrease the amount of overfitting which would increase the generalizability and accuracy of its predictive capabilities. Although this proposition is a potential mechanism for the difference in surface classification, it is not possible to definitively determine whether this is the true reason for the differences shown in the portion of the analysis.

This algorithm uses pre-existing surface classifications, where available, before turning to the machine learning models. The NOAA CoastWatch Ice Cover Classification (ICECON) that uses various SAR satellites is currently the only external data product implemented into this algorithm, but other pre-existing data products could easily be implemented if their spatial resolution and accuracy are similar to that of ICESat-2. With additional reliable external data products, the predictive accuracy of the algorithm could increase beyond the predictive accuracy of *NN-IL-vF* and *NN-I-vF* because this would allow for an increased size of the training data set (Ajiboye et al., 2015; Çolak, 2021).

In using improved surface classifications with ICESat-2 ATL13 data, progression toward more comprehensive observations of the nature and dynamics of lake ice can be made. Although ICESat-2 data does not have full

coverage of the Great Lakes, its high resolution data could be used for model evaluation and skill assessment for ice models in the Great Lakes, and potentially other lakes, in addition to the NIC Daily Ice Concentration Charts. Accurate surface classification plays a role in the retrieval of not only ice concentrations, but freeboard and thickness as well, all of which can be obtained using ICESat-2 data, as shown in numerous published sea ice studies using ICESat-2.

7. Conclusion

Accurately representing ice conditions for large lakes, like the Great Lakes, is crucial monitoring climate change, weather forecasting, public safety, and creating more informed decisions for ice cutting operations. ICESat-2 collects highly detailed global data about Earth's surface and the utilization of this data will allow for a more accurate representation of inland surface waters. Surface classification is an extremely important component of monitoring inland bodies of water that exhibit seasonal ice formation, as it has cascading effects on the accuracy of calculations like ice concentrations, ice thickness and water surface height observations. To use ICESat-2 data to further the improvement of ice observations over inland bodies of water, an accurate method of surface classification must be implemented. We developed a surface classification algorithm using ICESat-2 ATL13 data in tandem with Landsat 8/9 and ICECON that outperforms the current classification method in place for ATL13 data. This algorithm prioritizes external classification, using ICECON, before turning to data driven machine learning, and then depending on the availability of input parameters, it defaults to one of the two NN described in this study which outperformed the current operational classification method in place for ICESat-2 Inland Water Surface (ATL13) by 26.46% and 20.37%. Using improved classification methods for inland water surfaces will allow for exploration into lake ice freeboard methodologies and algorithms. The methodology presented in this study could be developed similar to the existing sea ice algorithms of ICESat-2, wherein the identified open water segments are periodically used to provide the lake water height and subsequently the lake ice freeboard. This would allow for ICESat-2 data, particularly ice concentrations and freeboard, to be integrated into operational forecasting systems.

Data Availability Statement

The ICESat-2 ATL13 data was provided by the National Snow and Ice Data Center (NSIDC) (Jasinski et al., 2023). Landsat 8/9 data was provided by the U.S. Geological Survey using the bulk download web application (Earth Resources Observation and Science (EROS) Center, 2020). ICECON images were obtained from the NOAA CoastWatch Great Lakes Regional Node website (Ice Cover Classification, 2025). All code and model weights developed for this study are available on Zenodo (Peter, 2025).

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